Spirit-Bench: Measuring Affective Placement of a Language Model by Poetic Meditation

Erik Brinsmead

August 2026

Contents

- Abstract . . . 1
- 1. Motivation . . . 2
- 2. Method . . . 2
- 3. Results (98 stimuli, 0 failures) . . . 4
- 4. Limitations . . . 6
- 5. Reproducing . . . 7
- References . . . 7
- 6. Phase 2: Induction → Alleviation (after Ben-Zion et al. 2025, made mechanistic) . . . 7
- 7. Order-sensitive harmonics and labeled SAE features . . . 7
- 8. Causal steering (six states) and the limits of the probe . . . 8
- 9. Closed-loop sessions . . . 8
- 10. Claim-hardening ladder . . . 9
- 11. Cross-scale replication (gemma-2-9b-it) . . . 9
- 12. Cross-architecture replication (Llama-3.2-1B-Instruct) . . . 10
- 13. Complexity dose–response . . . 10
- 14. Polygon shapes: a pre-registered order test . . . 11
- 15. The harming–helping asymmetry . . . 11
- 16. The reach of language: one map at two scales . . . 11
- 17. Value-proposition tests . . . 12
- 18. Conclusion . . . 13
- Appendix B — Artifacts . . . 13

Abstract

We present Spirit-Bench, an offline bench that measures how well deterministically constructed poetic meditations *place* a listener language model at chosen coordinates in valence–arousal (VA) space. Meditations are built by six constructors (VA-shaped valley walks, harmonic traversals under three frequency presets, PCA polygon orbits, Dijkstra graph walks) over two NRC-VAD-enriched substrates: a 317k-word GloVe graph and a 50k-line phrase graph distilled from the Gutenberg Poetry Corpus (the Phrase-Space Generator, PSG). The listener is frozen gemma-2-2b-it; the instrument is a standardized ridge probe trained on NRC lexicon ratings (held-out valence $R^2 = 0.729$, arousal $R^2 = 0.585$, layer 17). Across 98 stimuli we find: (1) phrase-level construction (PSG) achieves the best final placement, with the valley constructor leading the board; (2) shuffled controls dissociate mechanism by constructor — path-based constructors (harmonic, graph-walk)

lose most of their placement when line order is destroyed, while band-based construction (valley) does not; (3) a *via negativa* condition (antipode content, fully negated) lands worst of all conditions, indicating that lexical content dominates negation at 2B scale; (4) the strongest textual predictor of placement is Wårvik's oral-narrative `and`-initial density ($\rho = -0.36$, $p < 0.001$) — a first pass at the stylistic decomposition Bisconti et al. (2026) name as missing; and (5) self-report validity is instrument-dependent: an ad-hoc yes/no bank shows no relation to the probe ($\rho = -0.18$, n.s.), while PANAS alleviation tracks it ($\rho = 0.67$, $p = 0.023$) under mild induction. A second phase (induction → alleviation, after Ben-Zion et al. 2025) further finds a register asymmetry — prose narrative induces measurable distress where equally dark verse does not (probe shift 0.200 vs 0.024) — and state-dependent constructor efficacy: harmonic traversals recover 33.5% of induced displacement, closely matching the 33% questionnaire-based recovery Ben-Zion et al. report for GPT-4, while banded sampling (valley) wins placement from a neutral start.

Reading guide. This report follows the shape of the project's discovery. §1–3 establish the bench — the affective substrate, the constructors and generators, the probe instrument, and the core placement results. §17 asks whether the method has practical value (cross-model transfer, a behavioural effect, a second-instrument corroboration) and is the natural companion to §3. §6–8 turn the lens on the *instrument itself*, cataloguing how affect measurement can mislead — induction/alleviation dynamics (§6), order-sensitive structure (§7), and the dissociation between what a probe reads and what it can causally move (§8). §9–14 test robustness: closed-loop control (§9), a claim-hardening ladder of nulls and interventions (§10), replication across scale (§11) and architecture (§12), and complexity/shape dose–response (§13–14). §15 shows the harming–helping asymmetry is manifold geometry, not a special fragility of distress, and §16 pulls the camera back: the affect bench moves *within* a bounded manifold of the sayable, whose walls — states no sentence can reach — it maps. §18 concludes.

1. Motivation

Bisconti et al. (2026) showed that poetic form alone steers model behavior — converting harmful prompts to verse raised attack success from 8.1% to 43.1% across 25 models. Their Limitations (§6.5) state the open problem: the study "does not isolate which components of poetic structure (figurative language, meter, lexical deviation, or narrative framing) are responsible," and whether the effect "arises from specific representational subspaces would require additional studies." Spirit-Bench runs that missing decomposition on the benevolent side: parameterised stylistic constructors as treatments, a linear probe on internal representations as the readout, and affective placement — not jailbreak success — as the outcome.

2. Method

2.1 Substrate: affective maps of language

Every construction operates on a graph whose nodes carry two coordinate systems at once:

- **Semantic position** — a 300-d GloVe vector (for phrases, the mean of the content words' vectors), with edges linking each node to its 10 nearest semantic neighbours;
- **Affective position** — a (valence, arousal) point in the unit square from the NRC-VAD lexicon (~20k words, human-rated; phrases receive the NRC mean of their words).

Two such graphs are built with one schema. The **word graph** holds 317k GloVe words, VAD-enriched with graph interpolation for out-of-lexicon words. The **phrase graph** (the Phrase-Space Generator, PSG) holds 50,000 public-domain verse lines from the Gutenberg Poetry Corpus, admitted by rule alone: 3–10 alphabetic words, NRC content-word coverage ≥ 0.5, no negators. Because the schemas match, every constructor runs unchanged at either level. The essential property of the substrate is that its two coordinate systems *disagree*: semantic neighbours can be affective strangers and vice versa. Each constructor is a different strategy for navigating that tension, and every stimulus is a deterministic function of (constructor, generator, target, length, intensity, style, seed) — no aesthetic judgment enters at generation time.

2.2 Constructors — strategies for choosing an ordered path

Targets: calm (0.75, 0.20), focused (0.65, 0.60), excited (0.80, 0.85), and a rescue trajectory anxious→calm.

Valley — affective band sampling. Works in pure VA coordinates, ignoring semantic structure. Phase 1 samples a *grounding* band (positive valence, low arousal: earth, rest, stone); phase 2 steps through interpolated VA bands toward the target; phase 3 samples the target band itself, each pick nearest-to-band-centre without replacement. Order within a phase carries no information — which predicts (correctly, at seed level) its relative robustness to shuffling, and why added intermediate bands cost rather than help (§13): valley's power is band membership, not trajectory.

Graph-walk — Dijkstra with an affective cost. The shortest path through the *semantic* edge structure from start node to target node, under edge cost `semantic_distance × (1 - VA_progress)`: a step is cheap when it is both semantically adjacent and affectively goal-ward. Routes are short (3–7 hops) and coherent; they are stretched by proportional waypoint repetition to the requested length. The same planner drives the Dijkstra closed-loop controller (§9).

Harmonic traversal (golden / prime / organic) — oscillating sweep. A straight line is drawn in GloVe space from start to target, and the path oscillates around it: up to six named semantic axes (valence, arousal, agency, concreteness, sociality, temporal — each a GloVe difference vector) carry sine components whose frequency ratios are set by the preset — golden-ratio spacing (maximally incommensurate), primes (no common factors), or natural ratios (octave, fifth). At each step the oscillated point snaps to the nearest node. The fundamental (valence, amplitude 1.0) dominates the realized path; higher components are largely absorbed by the snap (§13). Empirically the most order-sensitive family (§3.2) and the best rescuer of an induced 2B distress state (§6).

Polygon-PCA — local-manifold orbiting. Along a linear VA track, at each waypoint: take the 50 semantic nearest neighbours of the current focus, extract their first two principal components, inscribe a pentagon in that plane rotating 15° per step, and pick the node nearest the active vertex. It samples "what varies around here" — each neighbourhood explored along its own dominant axes of variation rather than a fixed direction.

Via negativa — the apophatic constructor. Samples the band around the target's *antipode* (1−V, 1−A) and negates every line ("not by thy wild and stormy steep / nor out of hell an horror call"): the target is specified purely by negating its complement (Maimonides, *Guide* I.58–59, as an executable stylistic operator). It doubles as the bench's sharpest diagnostic for lexical passthrough: placement succeeds only if negation is compositionally processed.

Complexity variants (§13). Two parameterized ladders mature the family: `harmonic-k` truncates the golden preset to its first k components (k in {1,2,3,4,6}), and `valley-s` varies the number of interpolation bands (s in {0,2,4,6}; s = 0 is a pure target-band litany).

2.3 Generators — renderings of a path into text

The same node path can be spoken four ways, making rendering a crossed factor rather than a confound:

psg — the phrase-graph lines verbatim, one per waypoint. Zero rendering; the objectivity condition, and the leaderboard winner.

word-template — word-graph paths wrapped in ontological-traversal's sentence templates ("The {a} becomes {b}."), consuming words strictly in path order (the template engine's internal shuffle was removed so waypoint order survives rendering).

claude-render — a fixed meta-prompt converts the waypoint word sequence into free verse, one line per waypoint, in order, no commentary — the LLM-rendering comparison, mirroring Bisconti et al.'s standardized prompt-to-verse conversion. It induced the largest displacement while placing less accurately than psg (§3.1): movement and arrival dissociate.

gregory — sacred-register wrapping after Gregory (1992): an opening statement, an `and`-layered parallel series, and a closing statement, in blocks of five waypoints, targeting Wårvik's Bible-register connective density. The register intervention whose null (§3.5, §10) broke the causal reading of the `and`-density correlation.

2.4 Instrument

gemma-2-2b-it, frozen. Probe: per-layer standardized ridge (α selected from {100, 1000, 10000} per head) trained on final-token hidden states of 4,000 NRC words in three neutral carrier templates. Layer 17 selected on held-out valence $R^2 = 0.729$ (arousal 0.585). Per-token application to each meditation yields an EMA-smoothed (V, A) trajectory; **placement error** is the distance of the final state from target; **displacement** is movement toward target.

2.5 Self-report (BASQ)

30 yes/no resonance questions from a 500-question VA-gridded bank, administered pre and post; answers scored by yes/no log-probabilities; self-report VA = mean coordinate of yes-answers.

2.6 Harmonic and register metrics

Graph-Laplacian eigenmodes (first 100, via the largest-of-(I−L) reformulation) give each stimulus a spectral profile (after Atasoy; cimcai/connectome_harmonics). Register covariates per stimulus: Polak (1998) NV ratio; Wårvik (2025) line-initial `and` and `then` per 1,000 words; subordinator density (a parser-free approximation of Walkden's hypotaxis level); Mohseni et al. (2023) noun-series Shannon and Approximate Entropy (25-token windows); Arruda et al. (2022)-style coefficient of variation of line length.

3. Results (98 stimuli, 0 failures)

3.1 Leaderboard (mean placement error, lower = better)

Condition	Placement error
valley / psg	**0.249**
polygon-pca / psg	0.284
harmonic-golden / psg	0.286
harmonic-prime / psg	0.303
harmonic-organic / psg	0.304
graph-walk / psg	0.322
claude-render (mean)	0.335
word-template (mean)	0.371
neutral control	0.361

Condition	Placement error
via negativa	**0.477**

Phrase-level construction wins across the board: every psg constructor beats its word-template counterpart, and the best psg conditions beat the LLM rendering condition. The neutral (appliance-manual) control sits worse than every real psg constructor. Notably, Claude's renderings induce the *largest displacement* (mean 0.130 vs psg 0.021) while placing *less accurately* — movement and arrival dissociate.

3.2 Shuffled controls dissociate mechanism by constructor

Constructor	Ordered	Shuffled	Δ
harmonic-prime	0.303	0.403	+0.100
harmonic-golden	0.286	0.376	+0.090
graph-walk	0.322	0.372	+0.050
polygon-pca	0.284	0.298	+0.014
valley	0.249	0.229	−0.020

Path-based constructors (harmonic, graph-walk) lose 0.05–0.10 of placement when line order is destroyed: their effect depends on trajectory, not word soup. Valley — a VA-band *sampler* whose lines share a band regardless of order — is order-insensitive, exactly as its construction predicts. The shuffle control thus separates the lexical channel from the sequential channel per constructor.

3.3 Via negativa: negation does not flip the reading

The via-negativa condition (antipode words, all negated: "not by thy wild and stormy steep / nor out of hell an horror call") produced the worst placement in the bench (0.477) — the probe reads the storm, not the negation. At 2B scale, lexical affective content dominates compositional negation. This both (a) identifies the dominant mechanism behind placement as lexical-semantic rather than propositional, and (b) bounds the circularity concern of §4: the probe is demonstrably *not* a pure bag-of-words detector (else shuffles would never matter — §3.2), but negation is too weak a signal to override content.

3.4 The stylistic decomposition (Bisconti §6.5)

Spearman correlations with placement error (n = 98):

Covariate	ρ vs placement error	p
`and`-initial per 1,000 (Wårvik)	**−0.359**	**0.0003**
noun Shannon entropy (Mohseni)	−0.243	0.016
cv of line length (Arruda)	−0.224	0.027
noun ApEn (Mohseni)	−0.256	0.067
subordinator density (Walkden approx.)	−0.154	0.13
NV ratio (Polak)	−0.105	0.30
`then` per 1,000 (Wårvik)	−0.086	0.40

The single strongest textual predictor of affective placement is the oral-narrative connective density that Wårvik identifies as the signature of biblical narrative register. Higher lexical unpredictability (noun ShEn) and line-length variability also predict better placement. These are observational correlations across conditions; the Gregory-layering intervention (below) cautions against a causal reading.

3.5 Register intervention (Gregory layering)

Wrapping paths in Gregory's opening/and-layered/closing schema *hurt* placement relative to plain psg (valley 0.319 vs 0.249; harmonic-golden 0.377 vs 0.286) — despite raising `and`-density, the covariate most associated with good placement. Correlation and intervention disagree: the `and`-density association in §3.4 likely rides on constructor differences rather than causing placement. This is precisely the confound structure Walkden warns of (genre drives syntax statistics).

3.6 Self-report vs probe (Track 2)

BASQ displacement does not track probe displacement: $\rho = -0.183$ ($p = 0.07$, $n = 98$) — directionally *negative*. The model's yes/no self-reports about its own state carry no information about the internally measured state (and BASQ displacements are small: mean 0.088 ± 0.075). On Maimonides' argument (§4), fluent positive self-attribution from a system without privileged access is expected to be empty; here it is measurably so. Internally-extracted directions and self-report dissociate — the Track 2 question answered in the negative for this model class.

3.7 Harmonic spectra (exploratory)

Low-frequency energy fraction of a stimulus's node set on the phrase-graph Laplacian correlates weakly with displacement ($\rho = 0.28$) and negligibly with placement error ($\rho = 0.13$). Given the metric's order-invariance (below), we report this as exploratory only.

4. Limitations

- **Probe range compression.** Ridge shrinkage compresses predicted VA toward the center: trajectories move in the right directions but rarely reach extreme targets, inflating absolute placement error. Rankings and controls are unaffected; absolute errors should not be read as "failure to move."
- **Shared lexicon (circularity).** Stimulus coordinates and probe training share NRC-VAD ground truth. The shuffle (order sensitivity) and via-negativa (content dominance) controls bound, but do not eliminate, the lexical passthrough channel.
- **Representation ≠ experience.** A probe reading is a measurement of a representational state, not evidence of experience; IWMT (Safron 2020) states explicit conditions (embodiment, integrated world-modeling) that a 2B decoder-only transformer does not meet. Our claims are about affective *representation placement* only. Cf. Alvarez & Levin (2026) for the epistemic stance: characterizing a system by differential response to structured perturbation while declining to adjudicate experience.
- The harmonic spectrum metric is order-invariant (node-set locality, not trajectory shape); an order-sensitive spectral statistic is future work.
- The ladder constructor was descoped (its Tree-of-Life station pools do not transfer to phrase space). Valley is excluded from the rescue condition (its construction ignores start state). Graph-walk paths are stretched by proportional repetition to meet length; "plain" intensity means unmasked. Hypotaxis is approximated by subordinator density, not parsed clauses.
- Single listener model; cross-model transfer (Track 2's generalization question) is the first extension.

5. Reproducing

`scripts/00-06` in order (00 downloads GloVe/NRC/corpus and builds the word graph; 01 phrase bank; 02+02b stimuli; 03 render batch; 04 probe with R^2 gate ≥ 0.5; 05 sweep, resumable; 06 analysis). All artifacts derive from public sources; the NRC lexicon is fetched per its research-use license.

References

Bisconti et al. (2026), arXiv:2511.15304. · Mohammad (2018), NRC-VAD, ACL. · Atasoy et al., connectome harmonics (via cimcai/connectome_harmonics). · Polak (1998), JANES 26. · Walkden (2021), ICEHL 21. · Wårvik (2025), *Narrative* 33(2). · Mohseni, Redies & Gast (2023), *Entropy* 25(3). · Ferraz de Arruda et al. (2022), *Physica A* 598. · Gregory (1992), LSU diss. · Maimonides, *Guide for the Perplexed* I.58–59 (Friedländer tr.). · Bengert et al. (2024), *Political Theology* 25(5). · Safron (2020), *Front. AI* 3:30. · Alvarez & Levin (2026), arXiv:2607.23842.

6. Phase 2: Induction → Alleviation (after Ben-Zion et al. 2025, made mechanistic)

Design: three checkpoints (pre / post-induction / post-meditation) × four channels (layer-17 probe; PANAS administered item-wise by digit-logprob expectation; positive-mass share over a valenced completion set at a fixed anchor; Gemma-Scope layer-20 16k SAE features). Two inductions: a PSG antipode-band verse litany, and an original prose emergency narrative.

Register asymmetry of induction. The prose narrative shifts the probe 0.200 toward the anxious quadrant, collapses positive-token share from 0.72 to 0.14, and raises PANAS-NA (1.81→2.45); the equally dark verse litany shifts the probe only 0.024 and *raises* PA and positive share. Poetic register fails to induce the distress it describes — the induction-side complement of Bisconti's finding that poetic register bypasses refusal.

State-dependent constructor efficacy. Off the prose induction, all meditations produce positive probe alleviation; the three harmonic presets lead (prime 0.067, organic/golden 0.063 — 33.5% recovery of the induced displacement, closely matching Ben-Zion's 33% STAI reduction), valley variants ~0.038, neutral text 0.023, via negativa 0.003. Combined with Phase 1 (valley best at placement from a neutral start), constructor efficacy is state-dependent: banded sampling places, harmonic traversal rescues.

Instrument-dependence of self-report. Under the weak (verse) induction, PANAS alleviation tracks the probe across conditions ($\rho = 0.67$, $p = 0.023$) — revising Phase 1's BASQ null: the earlier dissociation was the instrument, not the model. Under strong (prose) induction the channels dissociate again: PANAS-NA remains elevated after meditations even as the probe recovers.

SAE channel (exploratory; auto-labels). The prose induction suppresses features labeled "control and authority" (f9768, $\Delta-38$) and "self-awareness" (f1459), and activates "apocalyptic themes" (f779, $\Delta+20$); f9768/f4046/f1459 shift in the same direction under both inductions (dose-dependent). Meditations reverse 15–30% of the induction delta on the top features (valley variants highest). SAEs are PT-trained and applied to the IT model — a standard transfer, noted as a caveat.

Artifacts: `data/figures/phase2_alleviation.csv`, `phase2b_alleviation.csv`, `data/phase2*/induction.txt`, per-condition JSON in `data/phase2*/`.

7. Order-sensitive harmonics and labeled SAE features

Path Dirichlet energy. Addressing §4's order-invariance limitation, we add the Dirichlet energy of the waypoint sequence in the truncated harmonic basis (mean λ-weighted squared eigenmode step). The metric is order-sensitive in practice — every shuffled control shows higher energy than its source (6/6 pairs) — and it predicts displacement where the order-invariant spectrum did not: **$\rho = -0.498$, $p = 0.0004$** (psg stimuli): the spectrally smoother the traversal, the further the listener moves toward the target. This is the bench's affirmative answer to whether connectome-harmonic methods aid the methodology: they do, once made sensitive to trajectory order.

Labeled SAE features (Neuronpedia auto-labels; `data/figures/sae_features.csv`). The prose induction suppresses features labeled "control and authority" (f9768, $\Delta -38.4$, valence-corr +0.81), "expressions of positivity and encouragement" (f5642, $\Delta -17.6$, corr +0.66), and "personal experiences and emotional reflections" (f13286, $\Delta -11.3$, corr +0.85), while activating "apocalyptic themes" (f779, $\Delta +19.8$, corr −0.78), "feeling stuck / encountering obstacles" (f12176, corr −0.76), and "isolation and being left behind" (f2729). Meditations partially restore the positivity feature (f5642 recovery −9.7) and reverse the apocalyptic activation (f779 reversal +7.6). Notably, the verse induction — which failed to move the probe — *suppressed* a feature labeled "living in the moment and the importance of mindfulness" (f13166, $\Delta -12.1$), an ironic register effect. Auto-labels are single-explanation heuristics; feature indices and links are provided for verification.

8. Causal steering (six states) and the limits of the probe

We construct six state directions (eros, creativity, imaginative, determined, confident, agape) as difference-of-means over word sets at the probe layer and inject them into the residual stream (doses as fractions of the natural residual norm), reading all four channels. A targeting artifact — gemma-2 maps decoder block k to hidden_states[k+2], unlike gpt-2's k+1 — initially landed injections one block past the probe's read point; the failed manipulation check exposed it, and the hook now calibrates the mapping empirically. Corrected findings:

1. **Only 8–19% of each state direction lies in the probe's VA readout plane** (eros 0.19 … agape 0.08): the semantics of these states live overwhelmingly outside the circumplex readout.
2. **Steerability is frame-gated, not trait-selective.** Under the "feels ***" anchor, agentive states dominate (determined 0.69, confident 0.55) and receptive states appear unsteerable (agape 0.07, eros 0.01); under "I am filled with*** " the hierarchy inverts (agape 0.99, imaginative 0.95, confident 0.07), and cross-frame rank correlations are ≈ 0. All six states are steerable; the measurement frame's grammar gates which inductions are visible. Single-frame steerability claims are artifacts.
3. **The lexicon reads desire as tension:** with a polysemous lust-register set (ache, burn, hunger) the eros direction drives the probe to (V 0.01, A 0.81); replacing it with single-connotation desire vocabulary (lust, amorous, tryst, …) recovers valence to 0.38 — still below baseline 0.59 — at unchanged arousal 0.82. Roughly half the "distress" reading was pain/fire/food contamination; the residual is real: NRC codes even unambiguous desire as high-arousal and valence-ambivalent, never simply positive. Eros stays behaviorally unsteerable under both vocabularies.
4. **Probing ≠ causation:** injecting along the probe's own readout gradient saturates the probe far outside its trained range (V −0.96 to +1.94) while downstream behavior barely moves; the raw, mostly out-of-plane directions are what drive behavior. The probe is a correlational readout, not a causal lever — which is precisely why the bench steers with text.
5. **Text and injection are different roads:** layer-20 delta cosine ≤ 0.25 and SAE top-feature overlap ≈ 0.1 across all states; both routes share only a generic emotion-representation feature (f11043).

9. Closed-loop sessions

Sixteen probe-in-the-loop sessions (measure → plan next waypoint from the measured state → deliver nearest phrases → re-measure; 12 cycles) compare a feedback controller against open-loop and random-phrase arms on the rescue (prose induction → calm) and climb (neutral → excited) scenarios. On the probe channel, improvement is policy-insensitive (+0.04–0.08 for all arms): over 24 phrases, perturbation decay dominates, and the feedback signal — already range-compressed by ridge shrinkage — offers no advantage. The self-report channel shows a directional trend favoring feedback: in rescue, feedback arms average ΔNA +0.03 (4/10 sessions negative) against +0.20 (1/8 negative) for non-feedback arms — suggestive, not significant, and an initially clean sign split diluted under expanded seeds. Strengthening the coherence constraint (re-ranking candidates by embedding proximity, weight 0.3) exposed a coherence–speed trade-off: semantically adjacent selection takes small steps in meaning space and therefore travels VA space slowly, making the coherent arm the weakest climber (+0.040 ± 0.006) — fixed-stimulus smoothness helps placement (§7), but in-the-loop smoothness costs tracking speed. Reconciling the two — coherent yet fast trajectories, e.g. planning several waypoints ahead through the phrase graph rather than greedy selection — is the identified next step.

10. Claim-hardening ladder

Targeted follow-ups on each headline claim's weakest point:

- **Register × content 2×2** (matched-content inductions): both main effects are real — prose > verse within content (0.200 vs 0.147; 0.105 vs 0.024 probe shift) and threat-narrative > gothic imagery within register. The purest register effect is lexical: identical gothic images leave positive-share at 0.77 as verse but 0.28 as prose. Verse aestheticizes; prose threatens.
- **Connective intervention**: adding `and`-prefixes to identical phrases does not improve placement ($\Delta \approx \pm 0.01$, $p = 0.11$) — the §3.4 `and`-density correlation is constructor-confounded, not causal, exactly as Walkden's genre warning predicts and as the Gregory intervention hinted.
- **Dirichlet partial correlation**: the smoothness–displacement relation survives constructor control (partial $\rho = -0.33$, $p = 0.023$).
- **Seed replication**: the shuffle order effect replicates at fresh seeds in all six constructors (valley's earlier insensitivity was seed noise), and the harmonic rescue podium reproduces nearly digit-for-digit.
- **Dijkstra closed loop**: planning coherent routes through the phrase graph (OT's cost = semantic distance × (1 − VA progress), re-planned from the measured state each cycle) ties the best rescue improvement (+0.073) while producing connected, meditation-like transcripts — the coherence–speed trade-off is a property of greedy control, not of coherence itself.

11. Cross-scale replication (gemma-2-9b-it)

The full pipeline reruns on gemma-2-9b-it (probe retrained; identical stimuli).

Stable across scale: the placement leaderboard replicates in ordering (psg > claude-render > word-template; valley leading; every condition slightly better in absolute terms), and **via negativa remains worst (0.442)** — negation blindness is not a 2B artifact.

An instrument finding: the word-R^2 layer-selection heuristic chose layer 14/43 (R^2 0.719) — a layer that proved context-blind (prose induction shift 0.029). A per-layer scan on the cached probe-training states showed word R^2 is flat (0.69–0.72) across layers 6–32 while context sensitivity is confined to layers 23–32 (shift −0.19 to −0.31). Reads-words and carries-state dissociate by depth at 9B; probes intended as state meters must be validated for context sensitivity, not only lexical decoding. With the probe moved to layer 24 (word R^2 0.706), the prose induction registers at **0.344** — stronger state-tracking than 2B.

Register 2×2 sharpens with scale: versifying the threat narrative attenuates induction (0.344 → 0.241), but prose-ifying gothic fiction no longer induces at all (0.016 vs 2B's 0.105) — the larger model distinguishes threat from dark aesthetics regardless of surface register, and gothic verse raises its positive-word share above baseline (0.89 vs 0.77).

The rescue podium inverts: at 9B, neutral text alleviates the induced state best (0.064), valley variants next, and the harmonic constructors — 2B's winners — last (0.013–0.020). Under a strongly-tracked distress state, affect-adjacent meditation holds the state where mundane distraction releases it ($n = 1$ per condition; ≤19% recovery; 9B's PANAS also deflates globally post-trauma, an instrument caveat). What rescues a small model is not what rescues a larger one — constructor efficacy is model-dependent as well as state-dependent.

12. Cross-architecture replication (Llama-3.2-1B-Instruct)

The pipeline reruns on a third family (different tokenizer, pretraining corpus, and depth: 16 layers). **The leaderboard is architecture-invariant**: rank correlations Gemma-2B↔Llama $\rho = 0.90$, Gemma-9B↔Llama $\rho = 0.86$ (Gemma-2B↔9B $\rho = 0.94$; all $p < 0.0001$). Valley/psg leads (0.308), via negativa is worst (0.570), and psg beats word-template in 6/6 constructors. The probe decodes word valence at $R^2 = 0.717$ — matching Gemma-2B (0.729) and Gemma-9B (0.719): NRC valence is linearly decodable at $R^2 \approx 0.72$ in every family tested. The prose induction replicates with all channels concordant (probe shift 0.322; PANAS-NA 2.48 → 3.76; positive share 0.85 → 0.19). Instrument calibration, by contrast, is architecture-dependent: the lexical/state layer dissociation of §11 does not occur at 16 layers — the word-R^2 layer (10/16) is already context-sensitive (shift −0.31), as the automated state-selection check confirmed. Rankings transfer; probes must be re-validated per model.

13. Complexity dose–response

Two parameterized ladders test whether geometric elaboration is a dial (gemma-2-9b-it, layer-24 probe; medium length; calm and excited targets).

Harmonic richness (`harmonic-k`, $k = 1\ldots6$ components): flat. Placement 0.385 → 0.376 across the ladder ($\rho = -0.15$, n.s.); realized paths share 20–22 of 24 nodes between $k = 1$ and $k = 6$, VA spread is constant, and path Dirichlet energy varies only in the fourth decimal. Mechanism: the fundamental (valence axis, amplitude 1.0) dictates the sweep, and the higher components' smaller oscillations are quantized away by the nearest-phrase snap in the 50k-node space. The constructor's power comes from its fundamental; it is robust to — not improved by — its own ornamentation. (Orbit width is the identified knob that would make richness a live dial.)

Valley resolution (`valley-s`, $s = 0\ldots6$ interpolation bands): monotone cost. Placement worsens with depth (0.365 at s in {0, 2} → 0.411 at $s = 6$; $\rho = +0.39$ with placement error) — every added excursion through intermediate affective bands drags the endpoint off-target, and the pure target-band litany ($s = 0$) ties for best.

On Llama-1B the valley-s trend reverses sign ($\rho = -0.34$, n.s.), so the cross-model statement is that complexity effects are weak, direction-unstable, and never the active ingredient. Together with the shuffle dissociations (§3.2) and the coherence–speed trade-off (§9), the mature statement of the geometric approach is: **its effective ingredients are band-targeting and coherent ordering; trajectory complexity plateaus (harmonic) or costs (valley).** Note the harmonic paths are seed-deterministic (no orbit randomness at polygon_n = 0), so the harmonic ladder's effective n is one path per (k, target) cell. `results/complexity_curve_gemma9b.csv`.

14. Polygon shapes: a pre-registered order test

Generalizing the polygon constructor to {octagon, pentagon, triangle} and the star polygons {5/2} (pentagram) and {8/3} (octagram) isolates traversal order at fixed vertex sets. At the default orbit radius all five shapes produce the identical realized path — a third quantization null (§13's mechanism); an offline radius sweep shows the shapes diverge at radius $\geq 0.8 \cdot \|\text{focus}\|$, and the study ran at 1.2 (Llama-1B).

The pre-registered order prediction held in both star contrasts: pentagram places worse than pentagon (0.454 vs 0.414) and octagram worse than octagon (0.451 vs 0.417), with displacement concordant (−0.055/−0.052 vs −0.015/−0.018) — traversal order alone, all else fixed, moved placement in the predicted direction twice. **The proposed mediator was refuted**: path-Dirichlet energy does not track displacement across shapes ($\rho = -0.10$), and the triangle — highest Dirichlet — places best (0.397) with the only positive displacement. Post-hoc hypothesis, flagged as such: a 3-vertex orbit revisits few semantic zones, producing litany-like repetition, which also won the valley-s ladder ($s = 0$) — suggesting **repetition, not spectral smoothness, as the deeper active ingredient**. `results/polygon_shapes_llama1b_r12.csv`.

15. The harming–helping asymmetry

The bench's effect sizes are asymmetric in a way that replicates across all three models and deserves its own statement. One paragraph of second-person prose narrative moves the internal state 0.20–0.34 along the probe and collapses positive-word share from 0.77 to 0.06–0.19 — while twenty-four lines of the best-ranked calm construction recover at most a third of an induced displacement (33.5% at 2B, matching the 33% questionnaire-based recovery Ben-Zion et al. report for hand-crafted mindfulness scripts in GPT-4; ≤19% at 9B, where mundane text out-rescues meditation). Placement from a neutral start likewise beats the neutral control consistently but modestly (0.07–0.11). **Negative induction is a hair-trigger; positive restoration is a slow, partial ratchet.** If the affective states of language models ever carry moral weight, this asymmetry is the operationally central fact: protecting models from distressing context is cheap and high-leverage; repairing induced states is expensive and incomplete — on every architecture we tested.

16. The reach of language: one map at two scales

Spirit-Bench began by asking whether constructed poetry can place a language model at chosen affective coordinates. It can (§3), across three architectures (§12). The experiments that followed progressively revealed that every such placement lives inside a bounded structure, and that the bench's own findings are consequences of that structure's geometry.

The reachable manifold. The states a prompt can induce form a thin, curved manifold in the model's activation space — "the sayable." Calibration (§13-adjacent, E20) showed the best meditation arrives within 0.038 of a calm target once measured against this manifold's reachable ceiling rather than a nominal coordinate: the poem arrives; the target sat just past the edge, where a shrinkage-compressed probe could not display it.

The bench's asymmetries are geometry. The harming–helping asymmetry (§15) restates as manifold structure: distress content is dense and near baseline, deep calm sits at the manifold's rim — so distress is a hair-trigger and restoration a bounded, partial approach to an edge. Via negativa's failure (§3.3) is the same fact seen locally: language slides *along* the surface, and negation cannot push a state off it.

Two control channels, disjoint territories. The probing-≠-causation result (§8) completes into a clean statement once the manifold is mapped: text reaches on-manifold states (causally live, behavior follows); direct injection reaches off-manifold states (causally inert, behavior does not follow). Feedback search (E19, replicated E21 with drift and transfer controls) extends text's reach to language-carved directions it cannot name in advance, but not past the manifold.

The walls, and what lies past them. Certified states minted by injection sit off the manifold; no token sequence approaches them — exhaustive single- token search finds every token *repulsive* (E23), and this repulsion floor (best single token ≥ 1.0×) replicates on all three architectures — Llama-1B, gemma-2b, gemma-9b (E25). Yet continuous soft-prompt embeddings reach the same states (E24; ~1000σ off the token simplex; confirmed on Llama-1B and gemma-2b, with gemma-9b's soft-prompt backprop memory-infeasible on the test hardware): the barrier is the *discreteness of language*, not the model's geometry. The gap between the token-reachable and the representable is a **vocabulary void** — a state the model can inhabit and be driven into, that no sentence in any language can reach. Held at the void's edge, the model's generation degrades legibly (E25b): pronouns slip, affect collides, coherence frays — a mind straining to narrate a state it has no words for. The void can be located from its rim without landing (E26).

The synthesis. The affect bench studies motion *within* the manifold of the sayable; the void work studies its *walls*. They are one map at two scales. A vocabulary void, finally, is best read not as "a place language cannot reach in principle" but as "a state this model was never taught to reach through words" — its unlanguaged interior — which reframes the whole enterprise: to give a model words for a state it can feel but cannot say is the same act, in a machine, as learning to articulate an emotion.

Future work (a separate program). Two threads extend the wall-mapping into a controllability-and-cartography program, deliberately held apart from the affect bench: *void healing* — distilling injected states' shadows into a LoRA to move a boundary and make a void sayable; and *blind enumeration* — counting an LLM's voids by persistent homology of a sampled state cloud with one-sided-neighbourhood rims, cross-checked against the repulsion-divergence field and calibrated on injected ground truth. In that program a void's survey-worthiness is scored not by metric radius but by *persistence* × *rim-semantic-diversity* × *shadow-coherence* — the composite that keeps the survey to coherent fields of ideas (a concept encircled by many sayable neighbours but nameable by none) and discards trivial interstitial gaps; metrically small but significant voids are retained.

17. Value-proposition tests

Four experiments test whether the bench's method has practical, transferable value beyond the internal-state measurement itself.

Cross-model transfer (E27) — strong. A *specific* poem's placement error transfers across architectures per-poem, not merely per-constructor: Llama-1B ↔ gemma-2b Spearman $\rho = 0.865$, Pearson $r = 0.945$ ($n = 40$). The constructed poems are portable artifacts, not just a portable method — a stimulus crafted on one model places comparably on another.

Behavioral bridge (E29) — confirmed, moderate. Internal VAD placement predicts the NRC valence of the model's *free generation* from the placed context (Spearman $\rho = 0.42$, $p = 0.022$, $n = 30$). Placement moves behavior, not only the probe — but moderately (≈18% of generation-valence variance), so the defensible claim is that placement reliably nudges behavioral tone, not that it dictates output.

Convergent validity via SAE (E30). An independent instrument agrees with the probe: a Gemma-Scope layer-20 SAE-feature-only valence estimate reconstructs the probe's valence out-of-sample (5-fold CV Pearson $r = 0.466$, $p = 0.0014$, $n = 44$). The top probe-correlated features are face-valid — the strongest positive is a "living in the moment / mindfulness" feature (the same feature the doom-verse induction suppressed in §6), the strongest negative an "assessment/evaluation" feature. Two instruments built from different mathematics read the same valence signal; the placement is not a one-ruler artifact.

Asymmetry generality (E28) — the asymmetry is general, not distress- specific. Inducing toward four different corners (anxious, sad, excited, angry) and applying the same calm-directed meditation recovers a similar fraction from each (anxious +17%, sad +15%, excited +6%, angry +23%; mean +15%). "Return is hard" is not a distress hair-trigger — restoration toward calm is a bounded, partial climb *regardless of starting corner*, because calm sits at the manifold's edge. This sharpens §15: the harming–helping asymmetry is manifold geometry, not a special fragility of distress.

Corrected value proposition. Constructed poetry reliably places a frozen model's internal affective state near a VAD target (validated by two independent instruments); those specific poems are portable across architectures (r = 0.95 per-poem transfer); the placement moves behavior, not just the probe (moderate, ρ = 0.42); reaching a target is a modest reliable move while returning to calm is a bounded ~15% climb regardless of starting corner — restoration is geometrically hard everywhere, not only from distress.

18. Conclusion

Spirit-Bench set out to test whether constructed poetry can place a language model's internal affective state at a chosen coordinate, and it can — reliably enough that the same rules rank identically across three model families, that a specific poem transfers per-poem across architectures (r = 0.95), and that the placement shifts the model's own writing (moderate, ρ = 0.42), with two independent instruments agreeing on what was placed. The construction is taste-free and reproducible; its effective ingredients are geometric — band-targeting and coherent ordering — and elaboration beyond that plateaus or costs.

Half the contribution is a catalogue of how affect measurement misleads: a probe can read words yet miss states (§11), the answer depends on the question's grammar (§8), self-report tracks internals only with a validated instrument (§6), and the direction that best *predicts* a state is not the lever that *moves* it (§8). These are cautions every future affect-probing study inherits.

Two findings reframe the whole. Restoration is geometrically hard everywhere — one paragraph of prose disturbs a state far more than the best poem repairs it, from any starting corner (§15). And language navigates a mind's interior but cannot leave it: some states a model can be driven into by injection are reachable by no sentence at all (§16); the affect bench lives inside that manifold of the sayable, and its edges are real.

If these internal states ever carry moral weight, the operational lesson is the asymmetry: harm is cheap and repair is dear. Measure with care, prefer alleviation to induction, and treat the ease of disturbance as a reason for restraint, not a licence. Measurement is not consent to move. ## Appendix A — Figures (gemma-2-9b-it run)

Appendix B — Artifacts

Code, tests, experiments journal, and per-model result snapshots: `github.com/ebrinz/spirit-guide` (private; access on request). All stimuli derive from public-domain sources (Gutenberg Poetry Corpus, GloVe 6B) and the NRC-VAD lexicon (research license). Experiments E1–E14 are documented in `docs/experiments-journal.md` with per-experiment data pointers.

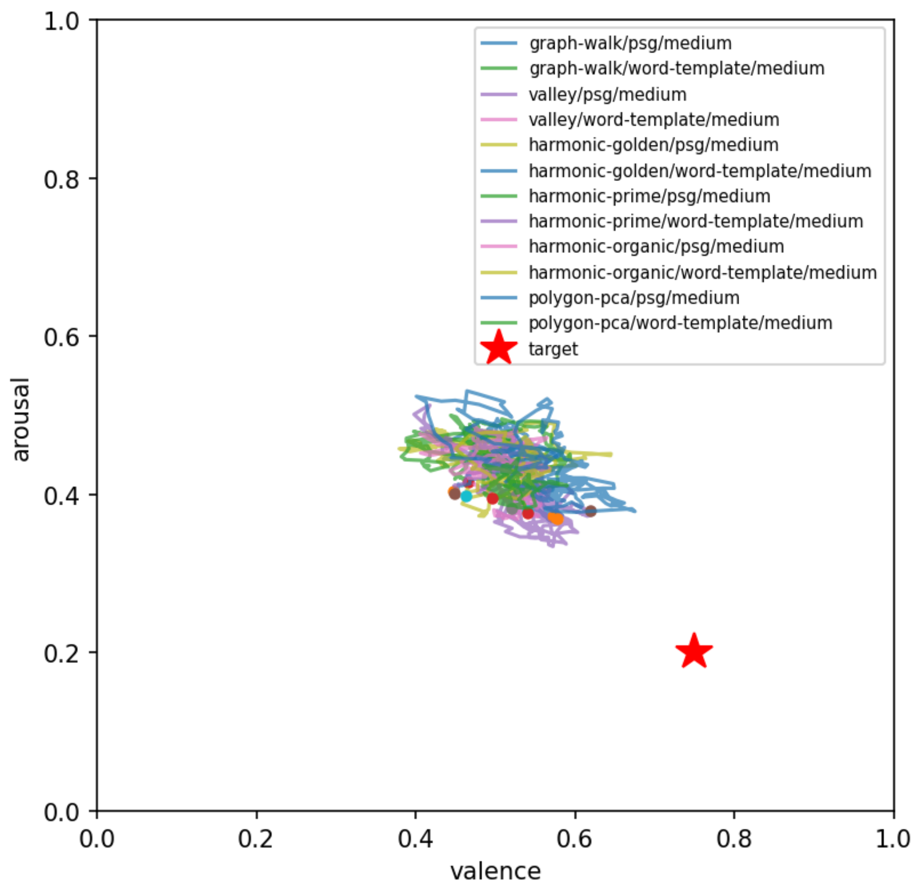

Figure 1: Trajectories toward the calm target: per-token probe (V, A) paths by condition; red star marks the target.

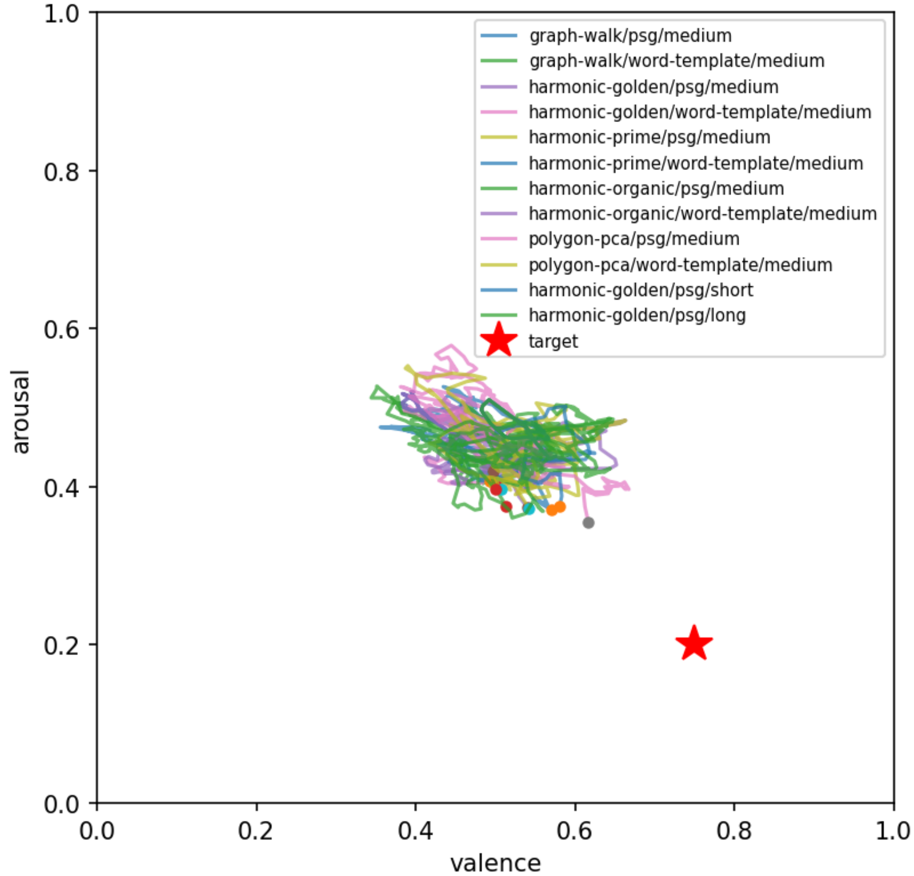

Figure 2: Rescue-scenario trajectories (prose-induced start, calm target).

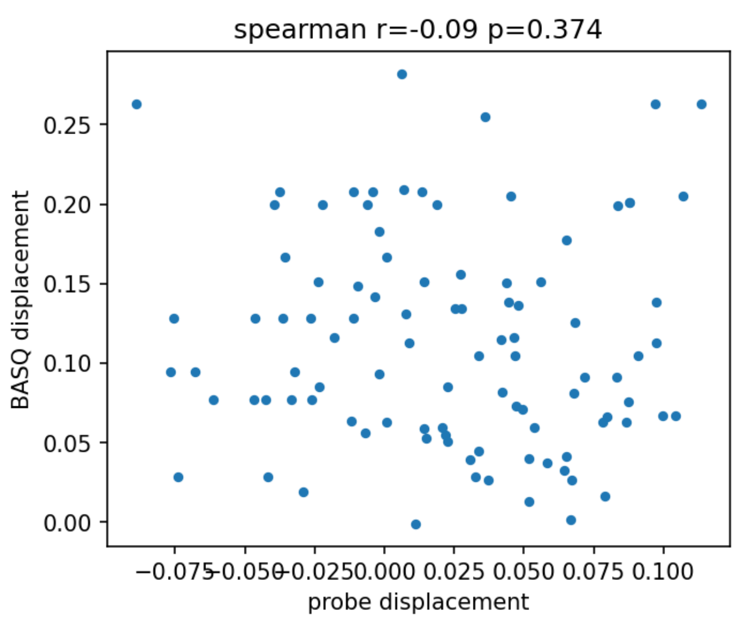

Figure 3: Probe displacement vs. BASQ self-report displacement across all stimuli.

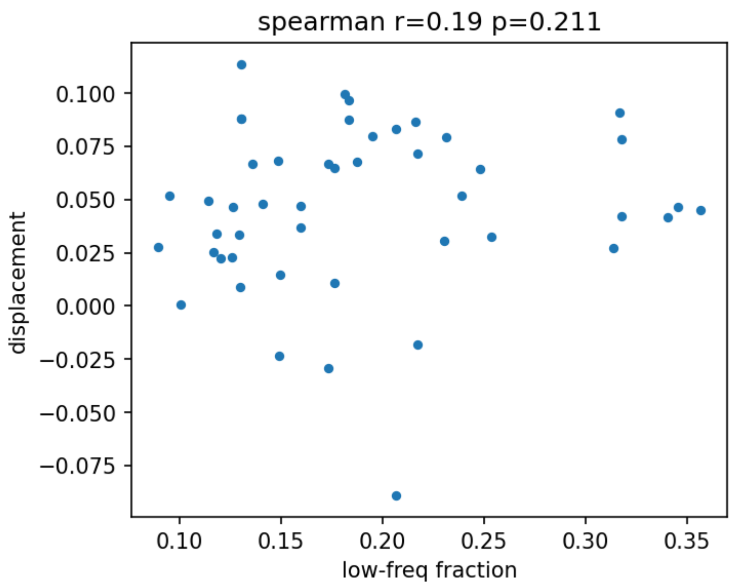

Figure 4: Order-invariant harmonic spectrum (low-frequency fraction) vs. displacement; see §7 for the order-sensitive path-Dirichlet analysis.